

Relative and Absolute File Paths

When you pass a simple filename like *pi_digits.txt* to `Path`, Python looks in the directory where the file that's currently being executed (that is, your *.py* program file) is stored.

Sometimes, depending on how you organize your work, the file you want to open won't be in the same directory as your program file. For example, you might store your program files in a folder called *python_work*; inside *python_work*, you might have another folder called *text_files* to distinguish your program files from the text files they're manipulating. Even though *text_files* is in *python_work*, just passing `Path` the name of a file in *text_files* won't work, because Python will only look in *python_work* and stop there; it won't go on and look in *text_files*. To get Python to open files from a directory other than the one where your program file is stored, you need to provide the correct path.

There are two main ways to specify paths in programming. A *relative file path* tells Python to look for a given location relative to the directory where the currently running program file is stored. Since *text_files* is inside *python_work*, we need to build a path that starts with the directory *text_files*, and ends with the filename. Here's how to build this path:

```
path = Path('text_files/filename.txt')
```

You can also tell Python exactly where the file is on your computer, regardless of where the program that's being executed is stored. This is called an *absolute file path*. You can use an absolute path if a relative path doesn't work. For instance, if you've put *text_files* in some folder other than *python_work*, then just passing `Path` the path `'text_files/filename.txt'` won't work because Python will only look for that location inside *python_work*. You'll need to write out an absolute path to clarify where you want Python to look.

Absolute paths are usually longer than relative paths, because they start at your system's root folder:

```
path = Path('/home/eric/data_files/text_files/filename.txt')
```

Using absolute paths, you can read files from any location on your system. For now it's easiest to store files in the same directory as your program files, or in a folder such as *text_files* within the directory that stores your program files.

NOTE

Windows systems use a backslash (\) instead of a forward slash (/) when displaying file paths, but you should use forward slashes in your code, even on Windows. The `pathlib` library will automatically use the correct representation of the path when it interacts with your system, or any user's system.

Accessing a File's Lines

When you're working with a file, you'll often want to examine each line of the file. You might be looking for certain information in the file, or